Markers for Automated Vehicles: Technologies, Applications, and Future Directions

1st Zirui Wu
Civil and Environmental Engineering
The Pennsylvania State University
State College, USA
zrwu@psu.edu

2nd Xianbiao Hu
Civil and Environmental Engineering
The Pennsylvania State University
State College, USA
xbhu@psu.edu

3rd Victor Jarlow
Research and Development
AstaZero
Göteborg, Sweden
victor.jarlow@ri.se

4th Timo Kero
Research and Development
AstaZero
Göteborg, Sweden
timo.kero@ri.se

5th Sean Brennan
Mechanical Engineering
The Pennsylvania State University
State College, USA
sbrennan@psu.edu

***Abstract*—Automated vehicles (AVs) heavily depend on sophisticated sensing technologies and advanced algorithms to accurately interpret their surroundings and make safe driving decisions. Despite significant advances in onboard sensors such as cameras, LiDAR, and radar, AV systems still encounter considerable challenges in complex real-world scenarios, including poor lighting, adverse weather conditions, occlusion, and densely populated urban environments. Consequently, external infrastructure solutions have emerged as vital components to enhance AV capabilities beyond onboard sensors alone. Markers, representing infrastructure-based technologies, offer distinct and stable reference points to assist vehicle perception, precise localization, accurate identification, and effective sensor calibration. This study systematically reviews three primary marker technologies: visual markers (e.g., QR codes, fiducial markers), magnetic markers embedded in road surfaces, and wireless tags (e.g., RFID, Bluetooth). For each marker type, we discuss operational principles, highlight key advantages, and analyze inherent limitations. Furthermore, we explore practical applications across various driving scenarios, from structured indoor facilities to challenging outdoor environments, while critically examining deployment challenges including infrastructure costs, environmental sensitivity, and signal interference. Finally, we outline several future research directions emphasizing multimodal marker integration, advanced sensor fusion techniques, and optimized deployment strategies, all aiming to further enhance the robustness and operational safety of automated vehicle systems.**

***Index Terms*—automated vehicle, vehicle perception, marker, localization, infrastructure support.**

I. Introduction

In recent years, the rapid development of automated vehicles (AVs) has significantly transformed transportation systems, promising increased safety, reduced congestion, and enhanced mobility. Despite substantial advancements in AV technology, including sophisticated sensors and artificial intelligence algorithms, automated driving systems continue to face challenges in accurately interpreting complex road environments, particularly under adverse conditions such as poor lighting, weather disturbances, or dense urban scenarios. These limitations underscore the importance of supplemental infrastructure solutions capable of extending the effectiveness of onboard sensing capabilities beyond current limitations.

Markers, serving as external infrastructure aids, provide distinct, easily detectable references to support various AV operational needs. They play critical roles in enhancing vehicles' localization accuracy, identifying critical environmental features, calibrating onboard sensors, and facilitating robust infrastructure communication. Common marker technologies employed include visual markers (e.g., QR codes and fiducial markers), magnetic markers embedded within road surfaces, and wireless tags (e.g., RFID and Bluetooth systems). Each marker technology exhibits unique advantages, such as cost-effectiveness, high precision, and reliable non-line-of-sight functionality. Conversely, each faces specific limitations, including environmental sensitivity, infrastructure complexity, and signal interference, necessitating careful assessment for their suitability across different operational scenarios.

The importance of integrating marker-based infrastructure into automated driving systems lies in its ability to address multiple operational challenges. Markers significantly enhance perception capabilities in degraded or complex environments—such as low-visibility conditions, occluded urban spaces, or GNSS-denied zones—thereby reducing dependence on vulnerable onboard sensors. They also offer reliable and precise localization verification, which is critical for executing fine maneuvers in densely populated urban areas or complex intersections. Furthermore, markers enable seamless integration of infrastructure-level information, allowing AVs to respond promptly to dynamic traffic conditions, unexpected hazards, and emergency scenarios. In addition to their functional contributions, marker technologies offer practical advantages that support scalable AV deployment. They are generally more cost-effective than advanced onboard devices, perform consistently across diverse environmental conditions, and exhibit high versatility—making them suitable for a wide

range of applications, from indoor parking and urban navigation to high-speed highway operations.

This study offers three key contributions to the existing body of knowledge. First, it systematically reviews and contrasts visual, magnetic, and wireless marker technologies, delineating their operational principles, strengths, and limitations. Second, it analyzes deployment challenges associated with each marker type and discusses viable solutions and practices. Finally, the paper identifies future research directions and emerging trends, emphasizing the integration and optimization of marker technologies to advance the robustness and scalability of AV infrastructure solutions.

II. Overview of Marker Technologies

Marker technologies serve as essential infrastructure components in AV systems, significantly enhancing vehicle perception, localization, and communication capabilities. This section provides an overview of three primary categories of markers: visual markers, magnetic markers, and wireless tags. Each type exhibits distinct characteristics, operational principles, and practical applications, addressing specific scenarios and challenges in AV deployment.

A. Visual Markers

Visual markers are physical patterns that can be optically detected and interpreted by vision-based systems. The most common types include barcodes, QR codes, and fiducial markers. Barcodes and QR codes store alphanumeric or binary information through arrangements of black-and-white elements. A barcode encodes data in a one-dimensional (1D) pattern, typically read by a laser scanner, while QR codes use a two-dimensional (2D) matrix to represent higher-density data. These markers are designed for quick recognition and decoding via computer vision algorithms under relatively constrained viewing conditions.

Fiducial markers, such as AprilTag [1], ArUco [2], and STag [3], are structured graphical patterns placed on flat surfaces. They are optimized not only for detection but also for pose estimation—calculating the relative position and orientation between the camera and the marker using geometric features. Most fiducial systems rely on contrast-rich patterns with high redundancy to ensure accurate localization even with partial occlusion or visual noise. These markers are particularly effective in environments where precise short-range localization is required.

The principal advantages of visual markers include low production and deployment costs, ease of integration into existing signage or infrastructure, and the ability to encode both identity and geometric information. However, their performance is highly sensitive to lighting conditions, visibility, viewing distance, and viewing angle. Issues such as motion blur, glare, shadowing, or environmental occlusion can substantially degrade marker detectability and accuracy [4]. Nevertheless, visual markers are widely used in indoor AV navigation, last-mile delivery systems, parking assistance, and calibration tasks.

B. Magnetic Markers

Magnetic markers operate by creating predictable variations in magnetic fields that can be detected by magnetoresistive or magneto-impedance (MI) sensors onboard AVs. These markers typically consist of permanent magnets (e.g., ferrite or neodymium) embedded beneath road surfaces in pre-defined patterns or intervals. When a vehicle passes over these markers, the onboard magnetic sensor array detects field perturbations caused by the markers, which are then processed to determine the vehicle's position relative to the roadway.

Modern systems such as the Global Magnetic Positioning System (GMPS) leverage high-sensitivity MI sensors to detect subtle magnetic field changes with millimeter-level precision [5]. Marker arrangements can be linear—positioned at regular intervals along lanes—or grid-based for complex areas like intersections. Magnetic signals are integrated with inertial measurements and odometry to estimate real-time pose.

The robustness of magnetic markers lies in their immunity to adverse environmental conditions. Since magnetic fields are unaffected by lighting, rain, fog, or dust, these systems provide highly reliable positioning, especially in GNSS-denied areas like tunnels or dense urban canyons. However, challenges persist, including infrastructure costs associated with embedding magnets, maintenance difficulties due to their in-road placement, and limitations in supporting bidirectional information exchange. Despite these, magnetic marker systems are already used in several real-world applications, such as autonomous buses and high-precision transit systems [6].

C. Wireless Tags

Wireless tags utilize electromagnetic communication to transmit identification and positional information to AVs without requiring visual contact. Two prevalent types are RFID (Radio Frequency Identification) tags and BLE (Bluetooth Low Energy) beacons.

RFID systems consist of a tag (transponder) and a reader (interrogator). Tags can be passive (powered by the reader's signal) or active (battery-powered). When a reader sends out an electromagnetic signal, nearby tags respond by backscattering or broadcasting stored data, which may include unique identifiers, geographic coordinates, or contextual metadata. UHF RFID systems, operating in the 860–960 MHz band, are particularly suitable for AV applications due to their read range (1–12 meters) and ability to function at high vehicle speeds [7].

BLE tags, often used in proximity detection and indoor positioning, emit low-energy signals that are received by BLE-compatible devices such as smartphones or onboard AV modules. These tags can support bidirectional communication and longer data payloads than RFID. BLE devices also allow for dynamic information broadcasting, such as real-time traffic updates or hazard warnings, making them versatile for infrastructure-to-vehicle (I2V) communication [8].

The major benefits of wireless tags include non-line-of-sight detection, large-scale deployment flexibility, and integration with digital infrastructure. However, their perfor-

mance can be degraded by electromagnetic interference, signal multipath effects, or physical obstructions in urban settings. Security and privacy concerns are also critical, as unauthorized access or spoofing of wireless signals could compromise AV behavior. Advanced encryption and authentication protocols are necessary to mitigate these risks [9]. Despite these issues, wireless tags are increasingly adopted in smart transportation systems, especially for vehicle identification, toll collection, and infrastructure-based guidance.

III. Applications

A. Localization

Accurate and reliable localization is foundational to the safe and efficient operation of AVs. It enables AVs to determine their precise position and orientation within the environment, which is critical for path planning, motion control, and decision-making. Conventional localization methods, such as Global Navigation Satellite Systems (GNSS), LiDAR-based simultaneous localization and mapping (SLAM), and visual odometry, often struggle in urban canyons, tunnels, and adverse weather conditions. Marker-based localization offers an infrastructure-supported solution that complements these onboard methods by providing robust and environment-invariant reference points.

1) Visual Marker-Based Localization: Visual markers enable precise localization by providing fiducial points with known spatial coordinates. When a vehicle-mounted camera detects a fiducial marker, it estimates the marker's pose relative to the camera using the Perspective-n-Point (PnP) algorithm. This transformation is then used to compute the vehicle's global pose. Single-camera systems are typically used for planar pose estimation, but they suffer from ambiguity under certain orientations and require a clear line of sight. Stereo camera setups improve robustness by resolving pose ambiguity and enabling geometric cross-validation between views [10].

Visual marker localization achieves sub-centimeter accuracy in controlled environments and is especially suited for indoor or structured spaces such as parking lots, warehouses, and intersections [11]. However, its performance degrades with partial occlusions, poor lighting, and distance between the marker and the vehicle. To improve resilience, researchers have proposed multi-modal fiducial markers that combine visual, thermal, and reflective components [12], ensuring reliable detection under diverse conditions.

2) Magnetic Marker-Based Localization: Magnetic markers offer a highly reliable alternative that operates independently of lighting and visibility. These markers are typically embedded in road surfaces and detected using magneto-impedance (MI) sensor arrays mounted on the vehicle. Each magnetic signature—often defined by a specific polarity pattern—provides a unique reference that can be matched to a pre-mapped database of known marker positions.

Localization using magnetic markers can be configured with either single or dual sensor arrays. While single arrays require additional estimation for heading using odometry, dual arrays directly estimate both position and orientation through geometric relations between front and rear detections. Systems like the GMPS integrate MI sensor readings with extended Kalman filters to achieve millimeter-level accuracy, even during tight maneuvers or curve tracking [6].

Despite their high accuracy and resilience, magnetic marker systems are infrastructure-intensive, requiring precise installation and maintenance of embedded markers. However, their reliability in GNSS-denied environments—such as tunnels, snow-covered roads, or urban intersections—makes them advantageous for certain high-precision applications.

3) Wireless Tag-Based Localization: Wireless tags, particularly RFID and BLE, support both absolute and relative positioning through radio-based detection. RFID tags are often embedded into pavement or roadside structures and contain encoded location data. When a vehicle-mounted reader detects a tag, it updates the vehicle's absolute position. Between tag detections, dead-reckoning using odometry or inertial measurement units maintains continuity. BLE-based localization relies on signal strength (RSSI), phase-of-arrival, or angle-of-arrival techniques to triangulate the vehicle's position relative to fixed beacons.

RFID-based localization offers reasonable accuracy (submeter in controlled scenarios) and operates well at high vehicle speeds [7] [9]. BLE tags, with longer communication ranges and bidirectional capabilities, provide flexibility in sparse deployment but are more susceptible to signal interference and latency issues [8]. To improve performance, fusion strategies incorporating sensor data and probabilistic estimation (e.g., particle filters or graph-based SLAM) are commonly employed.

Overall, marker-based localization serves as a reliable complement to onboard localization strategies, enhancing redundancy and robustness. It is particularly valuable in edge-case environments where traditional sensors may fail, and it enables scalable, infrastructure-supported navigation for future autonomous mobility systems.

B. Identification and Information Acquisition

Beyond localization, markers serve as effective tools for identification and contextual information retrieval in automated driving environments. Visual markers like QR codes can encode rich semantic content such as road sign types, speed limits, or temporary construction notices. These codes can be printed on traffic signs or road infrastructure and decoded by onboard cameras, enabling vehicles to confirm or supplement the information provided by perception systems [13].

Wireless markers, especially RFID tags, are particularly well-suited for object or location identification. When integrated into roadside infrastructure or other vehicles, RFID tags can transmit static or dynamic information—such as lane numbers, GPS coordinates, or hazard warnings—once detected by an AV's reader [7]. BLE beacons go a step further by enabling real-time data exchange, allowing vehicles to query infrastructure elements or receive time-sensitive alerts. These tags can be used in smart intersections to relay signal phase

TABLE I: Comparison of Different Marker Technologies

Category	Technology	Detection Range	Data Capacity	Advantages	Cost
Visual Marker	Barcodes	0.1-0.5m	8-20 characters	Simple, Cost-effective, Reliable	Very Low
	QR Codes	0.1-3m	4KB	High capacity, Error correction	Very Low
	Fiducial Markers	0.1-10m	Limited (ID only)	High accuracy positioning	Very Low
Magnetic Marker	Magnetic Markers	Contact-0.3m	0	Reliable	Low
Electronic Tag	RFID Tags	1-12m (Passive)	128B-8KB	No LOS needed, Mul-tiple reading	Medium
	BLE Tags	10-100m	255B/packet	Long range, Bidirectional	High

and timing (SPaT) data, enabling vehicles to anticipate red-light phases or coordinate complex maneuvers.

Marker-based identification is also useful in scenarios such as emergency response or work zones. Markers placed on emergency vehicles or temporary signage can broadcast their presence and operational status, allowing AVs to yield or reroute proactively. Markers can also be used in work zone geofencing for enhanced safety by transmitting boundary information [14]. This ability to integrate infrastructure-level context directly into the AV's decision-making process is essential for dynamic, real-world deployments.

C. Sensor Calibration

Marker technologies also play a crucial role in sensor calibration, which is fundamental to ensuring accurate data fusion and perception in multi-sensor AV systems. Calibration involves aligning the coordinate frames of different sensors, such as cameras, LiDAR, and inertial measurement units (IMUs), to enable coherent spatial interpretation.

Visual markers are widely used in sensor calibration due to their well-defined geometry and visual detectability. For instance, checkerboard patterns and AprilTags are employed to establish accurate 2D–3D correspondences between marker points and camera images [15]. For LiDAR sensors, specially designed markers (e.g., LiDARTags) that leverages different reflectivity patterns can be used for reliable LiDAR calibration [16].

Recent developments have proposed multimodal calibration targets that integrate visual and LiDAR features into a unified marker. One such system, A4LidarTag, uses laser-cut circular holes detectable in both image and point cloud domains to perform efficient extrinsic calibration [15]. These approaches enable semi-automated or fully automated calibration processes with high repeatability and low error.

IV. Deployment Challenges and Solutions

While marker technologies offer substantial benefits to AV operations, their real-world deployment faces a range of technical and operational challenges. These issues differ across marker types and affect scalability, reliability, and maintainability. This section categorizes and discusses key deployment challenges and corresponding mitigation strategies.

A. Visual Marker Challenges

The performance of visual markers is strongly influenced by environmental visibility and operational conditions, including lighting variations, shadows, glare, viewing angles, detection distance, and occlusion. QR codes require relatively normal viewing angles and good lighting, while fiducial markers may fail under motion blur or nighttime conditions [10]. In addition, pose ambiguity is a common problem in square planar fiducial markers, where two possible orientations may yield similar image projections, especially when the marker appears small in the image or is viewed from a distance [17].

To address these issues, several strategies have been proposed. Advanced image processing algorithms with gradient-based segmentation and edge detection can improve robustness against lighting variation. Active vision features, such as lighted markers and the integration of markers with overhead lighting, help mitigate lighting limitations. Marker designs such as STag and ArTuga incorporate thermal and LiDAR-detectable features to improve detection under adverse conditions [3] [12]. Furthermore, multi-view or stereo detection frameworks help resolve pose ambiguities by enforcing geometric consistency across observations. Robust pose selection techniques, such as reprojection error analysis and outlier rejection, further reduce the risk of incorrect localization.

Beyond standalone improvements, visual marker reliability can be enhanced through fusion with complementary modalities. For instance, visual markers provide long-range orientation cues but are vulnerable to long-range localization, while magnetic markers yield stable position references yet struggle with orientation sensitivity. Combining these modalities allows vehicles to exploit their complementary strengths.

B. Magnetic Marker Challenges

The main challenges associated with magnetic markers stem from infrastructure complexity and deployment cost. Embedding magnets in roadways requires precise installation and long-term durability, which can limit scalability. The accuracy of the magnetic localization system depends heavily on marker density and placement geometry. Sparse or inconsistent deployment can lead to degraded positioning accuracy, particularly in curves or intersections [6].

Solutions include optimizing marker spacing based on road geometry and vehicle kinematics. For instance, tighter spacing is recommended in curved sections, while wider spacing may be sufficient on straight road segments. Researchers have also proposed using magnetic grid patterns with alternating polarity (duo-placement) to improve uniqueness and reduce misidentification [18]. Integration with vehicle

odometry and inertial sensors through sensor fusion frameworks, such as extended Kalman filters, further enhances performance and error resilience. In addition, recent research explores passive RFID-embedded magnetic markers that can simultaneously provide identity and positioning information, improving system flexibility while minimizing cost [6].

While magnetic markers are resilient to weather and lighting, their performance is sensitive to external ferrous materials, fluctuating electric currents in nearby infrastructure, and vehicle height variations caused by road roughness or suspension motion. These factors can distort field measurements and reduce localization accuracy. Addressing these sensitivities requires careful sensor placement and calibration, as well as filtering techniques to suppress transient disturbances.

From a maintenance perspective, magnetic markers must be durable under heavy vehicular loads and environmental stress. Using ferrite-based magnets instead of neodymium reduces costs while maintaining adequate magnetic field strength. However, periodic inspections and recalibration are necessary to ensure continued accuracy and reliability.

C. Wireless Tag Challenges

Wireless tags face challenges related to signal reliability, electromagnetic interference, and cybersecurity. RFID and BLE systems are susceptible to multipath propagation and interference from metallic structures or electronic noise. In high-density environments, signal collisions and inconsistent read ranges can result in packet loss or erroneous localization [9]. Moreover, security risks such as spoofing, replay attacks, and unauthorized access pose potential threats to AV decision-making and user privacy.

To mitigate these challenges, robust system designs incorporate frequency diversity, anti-collision protocols, and error correction techniques. Advanced readers employ time-division multiple access (TDMA) or frequency-hopping strategies to minimize collision and enhance throughput. For security, encryption schemes, mutual authentication, and access control mechanisms are essential. Privacy-preserving designs, including the use of dynamic tag IDs, randomized response protocols, and limited-range broadcasts, help mitigate tracking risks and unauthorized surveillance.

Another important consideration is the power source for active tags. BLE beacons, while powerful, require periodic battery replacement or recharging. Passive RFID tags offer maintenance-free operation but have limited range and capacity. Hybrid systems that combine passive RFID with energy-harvesting technologies or low-power wake-up radios are emerging as promising solutions.

V. Discussion and Future Directions

A. Application Suitability

Due to the distinct characteristics of different types of markers, each is best suited for specific application scenarios and environments, as summarized in Table II. Visual markers are best suited for structured, low-speed environments such as parking lots or indoor facilities, where high-precision pose estimation and controlled lighting conditions prevail. Magnetic markers are ideal for outdoor applications with poor visibility or GNSS-denied scenarios like tunnels or snow-covered roads, offering reliable, weather-resilient performance. RFID tags enable accurate, high-speed localization in linear environments such as bridges and tunnels, while BLE tags support long-range communication and fleet-level monitoring through real-time data exchange. In complex urban settings, the combined use of RFID and BLE technologies facilitates seamless communication and adaptive navigation in dynamic, obstacle-rich environments

B. Future Directions

Despite significant progress in marker technologies, their integration into AVs is still limited. The following areas offer potential for future research.

1) Integration with Multiple Sensors: Current approaches primarily integrate identifiers with odometry for drift correction, which is insufficient under sensor degradation or in low-traction conditions. Future research should develop multi-sensor fusion frameworks that incorporate identifier data with perception sources such as visual SLAM, LiDAR, radar, and IMUs. Combining the precise, absolute positioning from identifiers with the rich environmental awareness of SLAM systems can enhance overall localization reliability. Probabilistic frameworks such as factor graph optimization and pose graph SLAM can address uncertainty and enable globally consistent estimation. Adaptive fusion strategies that dynamically weight sensor inputs based on context (e.g., lighting, weather, GNSS availability) will further improve robustness.

2) Advanced Algorithms for Real-Time Processing: Most current systems rely on linear models such as the Extended Kalman Filter, which are limited in highly dynamic or nonlinear environments. More advanced alternatives like Unscented Kalman Filters and non-parametric Bayesian filters can better capture non-linearities and complex dynamics. Machine learning techniques, particularly deep neural networks, offer potential for modeling high-dimensional input data and adapting to real-time changes. Hybrid architectures that combine learning-based and model-based approaches can strike a balance between flexibility and interpretability. Moreover, convex optimization and sequential convex programming can improve decision-making speed and accuracy in large-scale data fusion and motion planning tasks.

3) Optimization of Markers Deployment: The utility of markers depends heavily on strategic deployment. Heuristic or uniform placement can result in poor coverage and redundancy. Data-driven approaches, using traffic flow, road geometry, and environmental factors, can guide optimal marker density and positioning. Mixed-integer linear programming (MILP), genetic algorithms, and reinforcement learning could be explored for deployment optimization. Simulation environments such as digital twins can help assess visibility, coverage, and detection reliability under varied conditions. In addition, adaptive deployment—where markers are dynamically

TABLE II: Comparison of Marker Technologies for Various Scenarios

Scenario	Example	Recommended Marker	Reasons
Poor visibility/weather conditions	Outdoor environment with harsh weather	Magnetic Markers	Reliable detection in all environmental conditions.
Low-speed, high-precision positioning	Parking lot navigation	Fiducial Markers, Magnetic Markers	High positional accuracy for tasks like precision docking.
Long-range, high-speed positioning	Tunnel and bridge localization	RFID Tags, Magnetic Markers	Quick and robust detection.
Long-distance communication	Remote object tracking	BLE Tags	Extended detection range and bidirectional communication.
Complex urban environments	Workzones, intersections, urban streets	RFID Tags, BLE Tags	Effective without LOS and supports dynamic updates.

activated or repositioned—may offer scalable solutions for evolving infrastructure needs.

4) Integration of Multimodal Markers: Each type of marker—visual, magnetic, or wireless—has unique limitations that can be mitigated through integration. For example, visual markers can provide high-precision localization, magnetic markers ensure robustness under visual failure, and wireless tags add semantic data and long-range communication. Future research should develop cohesive frameworks for combining these modalities in real-time using context-aware fusion algorithms. Unified sensor modules that detect multiple marker types could reduce system complexity. Additionally, efforts toward standardizing marker encoding, communication protocols, and data formats will be essential to ensure compatibility across platforms and to enable broad deployment.

VI. Conclusion

This study systematically examined the use of markers in AV systems, highlighting their technologies, applications, and limitations. Visual markers provide high-precision localization at low cost but are constrained by visibility and lighting conditions. Magnetic markers offer reliable detection under adverse weather and in GNSS-denied environments, yet require infrastructure-intensive deployment. Wireless markers—such as RFID and BLE—support long-range, non-line-of-sight communication and dynamic data exchange, though they are sensitive to interference and infrastructure constraints.

While these markers have demonstrated value in specific scenarios, their integration into AV systems remains limited. Future research should prioritize multi-modal marker fusion, improved deployment strategies, and adaptive processing frameworks that leverage the complementary strengths of different marker types. Robust communication protocols and standardization will be essential to ensure interoperability and enable scalable, infrastructure-supported AV navigation.

References

[1] E. Olson, "AprilTag: A robust and flexible visual fiducial system," in *2011 IEEE International Conference on Robotics and Automation*, pp. 3400–3407, May 2011. ISSN: 1050-4729.

[2] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez, "Automatic generation and detection of highly reliable fiducial markers under occlusion," *Pattern Recognition*, vol. 47, pp. 2280–2292, June 2014.

[3] B. Benligiray, C. Topal, and C. Akinlar, "STag: A stable fiducial marker system," *Image and Vision Computing*, vol. 89, pp. 158–169, Sept. 2019.

[4] Y. Yan and H. W. Liu, "Research and Application of Encoding and Decoding Tech. of QR Code," *University of Science and Tech, Beijing*, 2015.

[5] K. Mohri, T. Uchiyama, L. V. Panina, M. Yamamoto, and K. Bushida, "Recent advances of amorphous wire CMOS IC magneto-impedance sensors: innovative high-performance micromagnetic sensor chip," *Journal of Sensors*, vol. 2015, no. 1, p. 718069, 2015.

[6] T. Ando, H. Mukumoto, K. Aoki, S. Okazaki, T. Nagao, H. Aoyama, M. Yamamoto, and K. Nakano, "Localization using global magnetic positioning system for automated driving bus and intervals for magnetic markers," *IEEE Transactions on Intelligent Vehicles*, vol. 8, pp. 878–887, Jan. 2023.

[7] J. Wang, D. Ni, and K. Li, "RFID-based vehicle positioning and its applications in connected vehicles," *Sensors*, vol. 14, pp. 4225–4238, Mar. 2014.

[8] Y. Zhuang, C. Zhang, J. Huai, Y. Li, L. Chen, and R. Chen, "Bluetooth localization technology: principles, applications, and future trends," *IEEE Internet of Things Journal*, vol. 9, pp. 23506–23524, Dec. 2022.

[9] R. Chen, X. Huang, Y. Zhou, Y. Hui, and N. Cheng, "UHF-RFID-based real-time vehicle localization in GPS-less environments," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, pp. 9286–9293, July 2022.

[10] R. Muñoz-Salinas, M. J. Marín-Jimenez, E. Yeguas-Bolivar, and R. Medina-Carnicer, "Mapping and localization from planar markers," *Pattern Recognition*, vol. 73, pp. 158–171, Jan. 2018.

[11] Z. Fang, Y. Chen, M. Zhou, C. Lu, N. Rottmann, R. Bruder, H. Xue, A. Schweikard, E. Rueckert, R. Nabati, *et al.*, "Marker-based mapping and localization for autonomous valet parking," in *IEEE/RSJ International Conference on Intelligent Robots and Systems WS (IROSWS)*, 2020.

[12] R. M. Claro, D. B. Silva, and A. M. Pinto, "ArTuga: A novel multimodal fiducial marker for aerial robotics," *Robotics and Autonomous Systems*, vol. 163, p. 104398, May 2023.

[13] J. Snyder, D. Dunn, J. Howard, T. Potts, and K. Hansen, "Invisible 2d bar code to enable machine readability of road signs—material and software solutions," *3M Transportation Safety Division, St. Paul, MN*, 2018.

[14] S. Thorén, L. Wikander, V. Jarlow, and T. Kero, "Model predictive geofencing for vehicle containment," in *2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 65–70, 2024.

[15] Y. Xie, L. Deng, T. Sun, Y. Fu, J. Li, X. Cui, H. Yin, S. Deng, J. Xiao, and B. Chen, "A4LidarTag: Depth-based fiducial marker for extrinsic calibration of solid-state lidar and camera," *IEEE Robotics and Automation Letters*, vol. 7, pp. 6487–6494, July 2022.

[16] J.-K. Huang, S. Wang, M. Ghaffari, and J. W. Grizzle, "Lidartag: A real-time fiducial tag system for point clouds," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4875–4882, 2021.

[17] G. Schweighofer and A. Pinz, "Robust pose estimation from a planar target," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, pp. 2024–2030, Dec. 2006.

[18] K. Ishii, K. Shimono, Y. Suda, T. Ando, T. Nagao, M. Yamamoto, and M. Segawa, "Vehicle Localization Using Magnetic Markers Incorporating EKF and Maximum Likelihood Estimation," *IFAC-PapersOnLine*, vol. 55, pp. 202–207, Jan. 2022.